Quantum Computing
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Weekly Exam Packets for Midterm 1 (lectures of August 25 – October 8, 2026)

One section per week. Read it after that week's review session. Each section starts with the results to have open in the scribes, then gives likely questions with full solutions. The questions are a guess at the exam, not questions from the professor: they take the exercises assigned in class and change the function, the numbers or one step. Every solution says which scribe results it uses. Numbers such as "Theorem 6, p. 8" refer to the scribes as submitted (`submission/Amatya_Scribes1.pdf`), the only copy allowed in the exam. A week only uses material up to that week. The few places that go beyond the scribes are marked.

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1 Week 1: Expectation and the Deterministic Search Lower Bound

Lectures of August 25 and 27. No exercises were assigned in these two lectures, so the questions below vary the theorems.

In the scribes	Result	Page
Deterministic search needs $n-1$ queries (adversary)	Theorem 1	3
Expectation	Definition 1	5
Linearity of expectation	Theorem 2	5
$\mathbb{E}[B(n,p)] = np$	Theorem 3	6
$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$	Theorem 4	6
$\mathbb{E}[G(p)] = 1/p$	Theorem 5	7

Problem 1.1 (Membership needs n queries)**.** The scribe shows that finding x needs $n-1$ queries when x is promised to be in the array. Show that deciding whether x is in the array at all needs n queries.

Solution. *Scribes:* Theorem 1, p. 3.

1. Fix a deterministic algorithm. The adversary answers every query with "not x".
2. Suppose the algorithm stops after at most $n-1$ queries. Then some slot j was never queried.
3. Two arrays agree with all the answers: one has x in slot j, the other has no x anywhere.
4. The algorithm saw the same answers on both, so it gives the same output on both. It is wrong on one.
5. So a correct algorithm makes n queries.

The promise is what saved one query on Aug 25: with a promise, $n-1$ answers of "no" leave only one possible slot. □

Problem 1.2 (Search with k copies of x)**.** An array of n slots contains x in exactly k slots, $1 \leqslant k < n$.

(a) Random sampling queries uniformly random slots (with replacement) until it finds x. What is the expected number of queries?

(b) Show that a deterministic algorithm needs $n-k$ queries in the worst case.

Solution. *Scribes:* Theorem 5, p. 7; Theorem 1, p. 3.

(a) Each query finds x with probability $p = k/n$, independently of the others, so the number of queries is geometric with $p = k/n$. Its expectation is $1/p = n/k$.

(b) The adversary answers "not x" to every query. Suppose the algorithm stops after $q < n-k$ queries and outputs a slot j. The array has $n-k$ slots without x, and only q of them have been used up by the answers, so at least one more slot without x remains. If j was queried, the answer at j was "not x" and the output is wrong. Otherwise the adversary puts the remaining empty slot at j and the k copies of x in other unqueried slots; this

agrees with all the answers, and the output is wrong. After $n-k$ answers of "no", the k unqueried slots all contain x, so $n-k$ queries suffice.

For $k=1$ this is the $n-1$ of the scribe. □

Problem 1.3 (Fixed points of a random permutation)**.** A permutation π of $\{1,\dots,n\}$ is chosen uniformly at random. Find the expected number of i with $\pi(i)=i$.

Solution. *Scribes:* Theorem 2, p. 5.

Let $X_i=1$ if $\pi(i)=i$ and 0 otherwise. By symmetry $\pi(i)$ is uniform on $\{1,\dots,n\}$, so $\mathbb{E}[X_i]=\mathbb{P}(\pi(i)=i)=1/n$. The number of fixed points is $X=X_1+\cdots+X_n$, and by linearity

$$\mathbb{E}[X]=\sum_{i=1}^{n}\frac{1}{n}=1.$$

The X_i are not independent (if $n-1$ points are fixed, so is the last one); linearity does not need independence. □

Problem 1.4 (A die, then coins)**.** Roll a fair die and let Y be the result. Then toss Y fair coins and let X be the number of heads. Find $\mathbb{E}[X]$.

Solution. *Scribes:* Theorem 3, p. 6; Theorem 4, p. 6.

Given $Y=y$, X is binomial $B(y,\frac{1}{2})$, so $\mathbb{E}[X\mid Y=y]=y/2$. So $\mathbb{E}[X\mid Y]=Y/2$, and by the tower rule

$$\mathbb{E}[X]=\mathbb{E}[\mathbb{E}[X\mid Y]]=\frac{\mathbb{E}[Y]}{2}=\frac{1}{2}\cdot\frac{1+2+\cdots+6}{6}=\frac{7}{4}.$$

□

Problem 1.5 (Two heads in a row)**.** A coin shows heads with probability p. Toss it in pairs: two tosses, then two more, and so on, until a pair is HH. Find the expected number of tosses.

Solution. *Scribes:* Theorem 5, p. 7.

A pair is HH with probability p^2, and pairs are independent. So the number of pairs is geometric $G(p^2)$, with expectation $1/p^2$. Each pair is two tosses, so the expected number of tosses is $2/p^2$. For a fair coin this is 8. □

2 Week 2: Coupon Collector, Tail Bounds, Birthday Paradox

Lectures of September 1 and 3.

In the scribes	Result	Page
Monte Carlo and Las Vegas algorithms	Definition 7	8
Coupon collector, $\mathbb{E}[X] = nH_n$	Theorem 6	8
$3n \log n$ queries access every slot w.h.p.	Theorem 7	9
Las Vegas search needs $\Omega(n)$ (Yao)	Theorem 8	9
Chernoff bounds (three forms)	Theorem 9	10
Exercise: no slot accessed fewer than $12 \log n$ times	Exercise 1	10
Markov's inequality	Theorem 10	11
Chebyshev's inequality	Theorem 11	11
Variance of the number of collisions	Claim 12	13
Exercise: birthday paradox, $m \geqslant 2\sqrt{n} + 1$	Exercise 2	13
Exercise: collision lower bound $k(k-1)/2n$	Exercise 3	14

Which inequality. Markov needs $X \geqslant 0$ and $\mathbb{E}[X]$: $\mathbb{P}(X \geqslant b) \leqslant \mathbb{E}[X]/b$. Chebyshev needs the variance: $\mathbb{P}(|X - \mathbb{E}[X]| \geqslant b) \leqslant \mathbb{V}[X]/b^2$. Chernoff needs a sum of independent 0/1 variables with mean μ:

$$\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant e^{-\mu\delta^2/3}, \qquad \mathbb{P}(X < (1-\delta)\mu) \leqslant e^{-\mu\delta^2/2} \quad (0 < \delta < 1),$$
$$\mathbb{P}(X \geqslant R) \leqslant 2^{-R} \quad (R \geqslant 6\mu).$$

For a statement about all n bins, slots or coupons, bound one of them and multiply by n (union bound).

2.1 Tail Bounds

Problem 2.1 (The same event, three bounds)**.** A fair coin is tossed n times and X is the number of heads. Bound $\mathbb{P}(X \geqslant 3n/4)$ with each inequality. Evaluate for $n = 100$.

Solution. *Scribes:* Theorem 10, p. 11; Theorem 11, p. 11; Theorem 9, p. 10.

$\mathbb{E}[X] = n/2$ and $\mathbb{V}[X] = n \cdot \frac{1}{2} \cdot \frac{1}{2} = n/4$.

1. Markov: $\mathbb{P}(X \geqslant 3n/4) \leqslant \frac{n/2}{3n/4} = \frac{2}{3}$.
2. Chebyshev: $X \geqslant 3n/4$ implies $|X - n/2| \geqslant n/4$, so the probability is at most $\frac{n/4}{(n/4)^2} = \frac{4}{n}$.
3. Chernoff: $3n/4 = \left(1 + \frac{1}{2}\right)\mu$ with $\mu = n/2$ and $\delta = \frac{1}{2}$, so the probability is at most $e^{-(n/2)(1/4)/3} = e^{-n/24}$.

For $n = 100$: 0.667, 0.04 and $e^{-4.17} \approx 0.016$. Markov does not improve with n, Chebyshev improves like $1/n$, and Chernoff improves exponentially. □

Problem 2.2 (n balls into n bins)**.** Throw n balls independently and uniformly into n bins, $n \geqslant 4$. Show that with probability at least $1 - 1/n^2$ no bin receives $3 \log_2 n$ balls or more.

Solution. *Scribes:* Theorem 9, p. 10.

1. Fix a bin and let X be its number of balls. X is a sum of n independent indicators, each 1 with probability $1/n$, so $\mu = 1$.
2. Take $R = 3\log_2 n$. For $n \geqslant 4$ we have $R \geqslant 6 = 6\mu$, so the third Chernoff form applies.
3. $\mathbb{P}(X \geqslant 3\log_2 n) \leqslant 2^{-3\log_2 n} = 1/n^3$.
4. Union bound over the n bins: $\mathbb{P}(\text{some bin has at least } 3\log_2 n \text{ balls}) \leqslant n \cdot \frac{1}{n^3} = \frac{1}{n^2}$. □

Problem 2.3 (Max load, upper tail, with the scribe's forms)**.** Random sampling makes $3n\ln n$ queries into an array of $n \geqslant 2$ slots. Show that with probability at least $1 - n^{-11}$ no slot is accessed $18\ln n$ times or more.

Solution. *Scribes:* Theorem 9, p. 10; Exercise 1, p. 10 for the lower tail.

1. Fix a slot and let Y be the number of queries that access it. Y is a sum of $3n\ln n$ independent indicators, each 1 with probability $1/n$, so $\mu = 3\ln n$.
2. $R = 18\ln n = 6\mu$, so the third form applies: $\mathbb{P}(Y \geqslant 18\ln n) \leqslant 2^{-18\ln n} = n^{-18\ln 2} \leqslant n^{-12}$, since $18\ln 2 \approx 12.48$.
3. Union bound over the n slots: the failure probability is at most $n \cdot n^{-12} = n^{-11}$. □

Beyond the scribes. The professor's notes state the sharper cap $9\ln n$. That needs the general Chernoff form $\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant \left(e^{\delta}/(1+\delta)^{1+\delta}\right)^{\mu}$ with $\delta = 2$, which the scribe does not state (proof in `practiceproblems.pdf`, Sections 10–11).

Problem 2.4 (Coupon collector with a different number of trials)**.** There are n coupons and each trial gives a uniformly random one. Bound the probability that some coupon is missing after

(a) $2n\ln n$ trials,

(b) $n\ln n + cn$ trials.

(c) What happens after only n trials?

Solution. *Scribes:* Theorem 7, p. 9; Theorem 2, p. 5.

A fixed coupon is missed by one trial with probability $1 - 1/n$, so it is missing after m trials with probability $\left(1 - \frac{1}{n}\right)^m \leqslant e^{-m/n}$.

(a) $e^{-2\ln n} = 1/n^2$ per coupon, so at most $n \cdot \frac{1}{n^2} = \frac{1}{n}$ by the union bound.

(b) $e^{-\ln n - c} = e^{-c}/n$ per coupon, so at most e^{-c} in total. For $c = 3$ this is about 0.05.

(c) The expected number of missing coupons is $n\left(1 - \frac{1}{n}\right)^n \approx n/e$, by linearity of expectation. About 37% of the coupons are still missing. □

Problem 2.5 (Boosting a Monte Carlo algorithm by majority vote)**.** An algorithm gives the correct yes/no answer with probability 2/3. Run it t times independently (t odd) and output the majority answer. How large must t be for an error of at most $1/n$?

Solution. *Scribes:* Definition 7, p. 8; Theorem 9, p. 10.

1. Let X be the number of correct runs: a sum of t independent indicators with $\mu = 2t/3$.
2. The majority is wrong only if $X < t/2$.
3. Write $t/2 = (1-\delta)\mu$: $(1-\delta)\cdot\frac{2t}{3} = \frac{t}{2}$ gives $\delta = \frac{1}{4}$.
4. Lower tail: $\mathbb{P}(X < t/2) \leqslant e^{-\mu\delta^2/2} = e^{-(2t/3)(1/16)/2} = e^{-t/48}$.
5. $e^{-t/48} \leqslant 1/n$ when $t \geqslant 48\ln n$.

So $\mathcal{O}(\log n)$ repetitions bring the error from $1/3$ down to $1/n$. □

Problem 2.6 (Empty bins)**.** Throw m balls independently and uniformly into n bins.

(a) Find the expected number of empty bins.

(b) Evaluate it for $m = n$, $m = n\ln n$ and $m = 2n\ln n$.

(c) For $m = 2n\ln n$, bound the probability that some bin is empty.

Solution. *Scribes:* Theorem 2, p. 5; Theorem 10, p. 11.

Let $Y_i = 1$ if bin i is empty and 0 otherwise, and let $Y = Y_1 + \cdots + Y_n$ be the number of empty bins. One ball misses bin i with probability $1 - 1/n$, and the balls are independent, so $\mathbb{E}[Y_i] = \mathbb{P}(Y_i = 1) = \left(1 - \frac{1}{n}\right)^m$.

(a) By linearity of expectation, $\mathbb{E}[Y] = n\left(1 - \frac{1}{n}\right)^m \leqslant ne^{-m/n}$, using $1 - x \leqslant e^{-x}$.

(b) For $m = n$: about n/e, so 37% of the bins are empty (for $n = 100$ the exact value is 36.6). For $m = n\ln n$: at most $ne^{-\ln n} = 1$. For $m = 2n\ln n$: at most $n \cdot n^{-2} = 1/n$.

(c) $Y \geqslant 0$, so Markov gives $\mathbb{P}(Y \geqslant 1) \leqslant \mathbb{E}[Y] \leqslant 1/n$.

The Y_i are not independent (if every other bin is empty, bin i is not), and linearity does not need them to be. Part (c) is the coupon collector bound again: Markov on the number of empty bins gives the same number as the union bound over the bins. □

Problem 2.7 (Las Vegas to Monte Carlo)**.** A Las Vegas algorithm $\mathcal{A}$ is always correct and makes T queries in expectation on every input.

(a) Build an algorithm that never makes more than $2T$ queries and fails with probability at most $\frac{1}{2}$.

(b) Get the failure probability down to $1/n$.

(c) Check (a) on random sampling for search.

Solution. *Scribes:* Definition 7, p. 8; Theorem 10, p. 11.

(a) Run $\mathcal{A}$ and stop it after $2T$ queries. If it has finished, output its answer, which is correct. If not, output "fail". Let X be the number of queries that $\mathcal{A}$ would make. $X \geqslant 0$ and $\mathbb{E}[X] \leqslant T$, so by Markov $\mathbb{P}(X \geqslant 2T) \leqslant \frac{\mathbb{E}[X]}{2T} \leqslant \frac{1}{2}$. The new algorithm fails only in that case.

Stopping after cT queries gives failure at most $1/c$ in the same way.

(b) Markov alone would need $c = n$, that is nT queries. Instead repeat the $2T$-query version $\lceil \log_2 n \rceil$ times with fresh randomness and stop at the first run that finishes. All runs fail with probability at most $\left(\frac{1}{2}\right)^{\log_2 n} = \frac{1}{n}$. The total is at most $2T\lceil \log_2 n \rceil$ queries.

(c) Random sampling with replacement finds x with probability $1/n$ per query, so the number of queries is geometric with mean $T = n$. Markov says that $2n$ queries fail with probability at most $\frac{1}{2}$. The exact value is $\left(1 - \frac{1}{n}\right)^{2n} \leqslant e^{-2} \approx 0.135$.

Markov is weak here, but it used nothing about $\mathcal{A}$ except its expected cost. Part (b) works because a Las Vegas run tells us whether it finished, so failures are detected and never output as answers. □

2.2 Lower Bound for Randomized Search

Problem 2.8 (Randomized search needs about $n/2$ expected queries)**.** Use Yao's principle to show that a Las Vegas algorithm for search makes at least $(n-1)/2$ expected queries on some input.

Solution. *Scribes:* Theorem 8, p. 9; Theorem 1, p. 3.

1. Input distribution μ: x is in a uniformly random slot, each with probability $1/n$.
2. Take a deterministic algorithm that is always correct. As long as every answer is "no", its queries follow one fixed order $q_1, q_2, \ldots$
3. It cannot stop while two or more slots are unqueried and x has not been seen (the adversary argument). So if x is in slot q_i with $i \leqslant n-1$, it makes at least i queries, and if x is in the last slot it makes $n-1$.
4. The expected number of queries under μ is at least
$$\frac{1}{n}\left(\sum_{i=1}^{n-1} i + (n-1)\right) = \frac{(n-1)(n+2)}{2n} \geqslant \frac{n-1}{2}.$$
5. By Yao's principle the expected cost of a randomized algorithm on its worst input is at least the expected cost of the best deterministic algorithm under μ. So it is at least $(n-1)/2 = \Omega(n)$. □

2.3 Birthday Paradox and Collisions

There are m people (queries) and n days (values), and X is the number of colliding pairs:

$$\mathbb{E}[X] = \frac{m(m-1)}{2n}, \qquad \mathbb{P}(X = 0) \leqslant \frac{\mathbb{V}[X]}{\mathbb{E}[X]^2} \leqslant \frac{1}{\mathbb{E}[X]},$$

$$\mathbb{P}(\text{no collision}) = \prod_{i=0}^{m-1}\left(1 - \frac{i}{n}\right) \leqslant e^{-m(m-1)/(2n)}.$$

Problem 2.9 (The numbers for $n = 365$)**.**

(a) How many people make the expected number of collisions at least 1?

(b) How many make the probability of a collision at least $\frac{1}{2}$, by the chain-rule bound?

(c) How many does the Chebyshev proof need to guarantee probability $\frac{1}{2}$?

Solution. *Scribes:* Exercise 2, p. 13.

(a) We need $m(m-1) \geqslant 2n = 730$. Since $27 \cdot 26 = 702$ and $28 \cdot 27 = 756$, the answer is $m = 28$, with $\mathbb{E}[X] = 756/730 \approx 1.04$.

(b) We need $e^{-m(m-1)/730} \leqslant \frac{1}{2}$, that is $m(m-1) \geqslant 730 \ln 2 \approx 505.997$. Since $22 \cdot 21 = 462$ and $23 \cdot 22 = 506$, the answer is $m = 23$. The exact product for $m = 23$ is 0.4927, so a collision has probability 0.507.

(c) Chebyshev gives $\mathbb{P}(X = 0) \leqslant 1/\mathbb{E}[X]$, so we need $\mathbb{E}[X] \geqslant 2$, that is $m(m-1) \geqslant 1460$. Since $38 \cdot 37 = 1406$ and $39 \cdot 38 = 1482$, the answer is $m = 39$. The condition $m \geqslant 2\sqrt{n}+1 \approx 39.2$ of the exercise gives $m = 40$.

The three answers differ (28, 23, 39) but all are about $\sqrt{n}$. Part (a) alone proves nothing about the probability: $\mathbb{E}[X] \geqslant 1$ does not imply that $\mathbb{P}(X \geqslant 1)$ is large. That step needs more than the expectation: either the direct product of part (b), or the variance, which is what the exercise uses with Chebyshev. □

Problem 2.10 (Why Chebyshev and not Chernoff?)**.** X is a sum of indicators X_{ij}. Why does the birthday proof not use the Chernoff bound? Why is Markov not enough?

Solution. *Scribes:* Claim 12, p. 13; Exercise 2, p. 13.

1. Chernoff needs mutually independent indicators. The X_{ij} are not: if persons 1 and 2 share a birthday and persons 2 and 3 share a birthday, then persons 1 and 3 do too. So $\mathbb{P}(X_{12} = X_{23} = X_{13} = 1) = 1/n^2$, not $1/n^3$.
2. They are pairwise independent: for two different pairs, $\mathbb{P}(X_{ij} = X_{kl} = 1) = 1/n^2 = \mathbb{P}(X_{ij} = 1) \cdot \mathbb{P}(X_{kl} = 1)$, whether the pairs share a person or not.
3. Pairwise independence is exactly what the variance needs. The cross terms $\mathbb{E}[X_{ij}X_{kl}] - \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}]$ are all zero, so $\mathbb{V}[X]$ is the sum of the $\mathbb{V}[X_{ij}]$, which is at most $\mathbb{E}[X]$. Chebyshev then applies.
4. Markov gives $\mathbb{P}(X \geqslant 1) \leqslant \mathbb{E}[X]$. That is an upper bound on the probability of a collision, and we want a lower bound.

The rule: Markov needs only $X \geqslant 0$, Chebyshev needs the variance (pairwise independence is enough), and Chernoff needs mutual independence. □

Problem 2.11 (The collision lower bound with numbers)**.** Let f be a random function into $n = 1{,}000{,}000$ values.

(a) Bound the probability that 500 queries find a collision.

(b) In general, bound it for $k = \sqrt{n}/2$ queries.

(c) How many queries are needed for probability $\frac{1}{2}$, and how many are enough?

Solution. *Scribes:* Exercise 3, p. 14; Exercise 2, p. 13.

By the Sep 3 exercise, k queries find a collision with probability at most $k(k-1)/(2n)$. The reason is that query number s hits one of the $s-1$ earlier answers with probability at most $(s-1)/n$, and the union bound adds these up.

(a) $\frac{500 \cdot 499}{2{,}000{,}000} \approx 0.125$.

(b) $\frac{k(k-1)}{2n} < \frac{k^2}{2n} = \frac{n/4}{2n} = \frac{1}{8}$, for every n.

(c) Needed: $\frac{k(k-1)}{2n} \geqslant \frac{1}{2}$ means $k(k-1) \geqslant 1{,}000{,}000$. Since $1000 \cdot 999 = 999{,}000$ is too small, $k \geqslant 1001$. Enough: by the birthday exercise, $m \geqslant 2\sqrt{n} + 1 = 2001$ random queries collide with probability at least $\frac{1}{2}$.

So the answer lies between 1001 and 2001 queries: $\Theta(\sqrt{n})$. These are two separate proofs. The upper bound analyses one algorithm (random queries, Chebyshev). The lower bound holds for every algorithm, adaptive or randomized (union bound). □

3 Week 3: Leader Election, Law of Large Numbers, Random Walks

Lectures of September 8 and 10.

In the scribes	Result	Page
Exercise: no deterministic leader election (symmetry)	Exercise 4	15
Exercise: at least one leader, $\geqslant 1 - 1/n^2$	Exercise 5	17
Two leaders form a bad pair	Claim 13	17
Two referee sets rarely miss each other	Claim 14	17
Exercise: more than one leader, $\leqslant 2\log^2 n/n$	Exercise 6	17
Messages and rounds	Theorem 15	18
Exercise: weak law from Chebyshev	Exercise 7	19
$p_{t+1} = p_t P$	Lemma 17	20
Exercise: pP is a probability vector	Exercise 8	21
$\pi_v = d(v)/2m$	Theorem 18	22
Convergence iff connected and not bipartite	Theorem 19	22

3.1 No Deterministic Leader Election

Problem 3.1 (The symmetry argument for $n = 3$)**.** Three anonymous nodes form a complete network, and all start in the same state.

(a) Write down the port numbering of the scribe's proof for $n = 3$, and say on which port a message sent on port 1 arrives.

(b) Run the deterministic algorithm "in round 1, send `hi` on port 1; a node that receives `hi` on port 2 becomes leader" on it.

(c) Which assumptions of the proof does the randomized algorithm of Sep 8 not satisfy?

Solution. *Scribes:* Exercise 4, p. 15.

(a) Port j of node i leads to node $i + j \bmod 3$:

Node	Port 1 leads to	Port 2 leads to
0	1	2
1	2	0
2	0	1

The edge between i and $i + 1$ is port 1 at i and port 2 at $i + 1$, so a message sent on port 1 arrives on port $3 - 1 = 2$.

(b) Every node sends `hi` on port 1, and by (a) every node receives exactly one `hi`, on port 2. So all 3 nodes become leaders. In general the proof shows that all nodes are always in the same state, so the number of leaders is 0 or n.

(c) The proof needs two things: all nodes start in the same state, and the algorithm is deterministic. The Sep 8 algorithm assumes unique ids, so the nodes do not start in the same state. It is also randomized: each node flips its own coins (whether to be a candidate, which referees to contact), so the induction step "the algorithm is deterministic, so they move to the same state" fails. □

3.2 Leader Election with a Changed Parameter

Each node becomes a candidate with probability p, and each candidate contacts k random referees; in class $p = 2\ln n/n$ and $k = \left\lceil \sqrt{n \ln n} \right\rceil$. A changed parameter only changes the numbers in four estimates (R_u is the referee set of u):

$$\mathbb{P}(\text{no candidate}) = (1-p)^n \leqslant e^{-pn}, \qquad \mathbb{P}(R_u \cap R_v = \emptyset) \leqslant \left(1 - \frac{k}{n}\right)^k \leqslant e^{-k^2/n},$$

$$\mathbb{P}(\text{more than one leader}) \leqslant \binom{n}{2} p^2 \, e^{-k^2/n}, \qquad \mathbb{E}[\text{messages}] \leqslant 2k \cdot np.$$

Problem 3.2 ($p = 3\ln n/n$)**.** Redo the analysis with candidate probability $p = 3\ln n/n$ and the same k.

Solution. *Scribes:* Exercise 5, p. 17; Claim 14, p. 17; Exercise 6, p. 17; Theorem 15, p. 18.

1. At least one leader: the candidate with the smallest id always wins at all its referees, so there is no leader only if there is no candidate. $\mathbb{P}(\text{no candidate}) = (1-p)^n \leqslant e^{-pn} = e^{-3\ln n} = 1/n^3$.
2. Two candidates with a common referee cannot both be leaders. Two referee sets are disjoint with probability at most $e^{-k^2/n} \leqslant e^{-\ln n} = 1/n$, as before.
3. A fixed pair is bad (both are candidates, with no common referee) with probability at most p^2/n. The expected number of bad pairs is at most
$$\binom{n}{2}\frac{p^2}{n} \leqslant \frac{n^2}{2} \cdot \frac{9\ln^2 n}{n^2} \cdot \frac{1}{n} = \frac{9\ln^2 n}{2n}.$$
4. By Markov, $\mathbb{P}(\text{more than one leader}) \leqslant \mathbb{P}(\text{at least one bad pair}) \leqslant \frac{9\ln^2 n}{2n}$.
5. Messages: at most $2k$ per candidate and $3\ln n$ candidates in expectation, so at most $6k\ln n = \mathcal{O}\left(\sqrt{n}\log^{3/2} n\right)$.

More candidates make "no leader" less likely, and "two leaders" and the message count slightly larger. The asymptotics do not change. □

Problem 3.3 (Only $\sqrt{n}$ referees)**.** What breaks if $k = \lceil\sqrt{n}\rceil$, without the logarithm?

Solution. *Scribes:* Claim 14, p. 17; Exercise 6, p. 17.

1. Now $k^2/n \approx 1$, so two referee sets are disjoint with probability about $e^{-1} \approx 0.37$ instead of $1/n$.
2. The bound on the expected number of bad pairs becomes $\frac{n^2}{2} \cdot \frac{4\ln^2 n}{n^2} \cdot e^{-1} = \frac{2\ln^2 n}{e}$, which is larger than 1. Markov says nothing.
3. The algorithm itself fails, not only the proof. Take the two candidates with the smallest ids. With probability about 0.37 they share no referee. Then, unless the

second one happened to pick the first as a referee, none of its referees hears a smaller id, and both declare themselves leader.

This is the birthday paradox: two random sets of size k out of n meet reliably only when k^2 is larger than n by a log factor. □

Problem 3.4 (Too few candidates)**.** What happens with $p = 1/n$?

Solution. *Scribes:* Exercise 5, p. 17.

The expected number of candidates is 1, and $\mathbb{P}(\text{no candidate}) = \left(1 - \frac{1}{n}\right)^n$, which tends to $1/e \approx 0.37$. So with constant probability nobody is elected. The factor $\ln n$ in p is what pushes this probability down to $1/n^2$. □

Problem 3.5 (Numbers for $n = 1{,}000{,}000$)**.** Evaluate the class algorithm for $n = 10^6$.

Solution. *Scribes:* Exercise 6, p. 17; Theorem 15, p. 18.

$\ln n \approx 13.82$.

1. Expected candidates: $np = 2 \ln n \approx 27.6$.
2. Referees per candidate: $k = \left\lceil \sqrt{10^6 \cdot 13.82} \right\rceil = 3717$.
3. Expected messages: at most $4k \ln n \approx 205{,}000$. The trivial algorithm, where everyone sends its id to everyone, uses about $n^2 = 10^{12}$.
4. Failure probability: at most $\frac{1}{n^2} + \frac{2 \ln^2 n}{n} \approx 10^{-12} + 3.8 \times 10^{-4}$.
5. Rounds: 3. □

3.3 Law of Large Numbers

Problem 3.6 (Estimating the bias of a coin)**.** A coin shows heads with unknown probability p. We toss it n times and estimate p by the fraction of heads. How large must n be so that the estimate is within ε of p with probability at least $1 - \delta$? Evaluate for $\varepsilon = \delta = 0.05$.

Solution. *Scribes:* Exercise 7, p. 19.

Let $X_i = 1$ if toss i is heads. Then $\mathbb{E}[X_i] = p$ and $\mathbb{V}[X_i] = p(1-p) \leqslant \frac{1}{4}$, because $p(1-p)$ is largest at $p = \frac{1}{2}$. The estimate is the average $\overline{X}_n$ of $X_1, \ldots, X_n$. By the weak law (the Sep 8 exercise, which is Chebyshev applied to the average),

$$\mathbb{P}\Big(\Big|\overline{X}_n - p\Big| \geqslant \varepsilon\Big) \leqslant \frac{\sigma^2}{n\varepsilon^2} = \frac{p(1-p)}{n\varepsilon^2} \leqslant \frac{1}{4n\varepsilon^2}.$$

This is at most δ when $n \geqslant \frac{1}{4\varepsilon^2\delta}$. For $\varepsilon = \delta = 0.05$: $n \geqslant \frac{1}{4 \cdot 0.0025 \cdot 0.05} = 2000$ tosses.

The bound does not need p, which is the point: p is what we are trying to learn. The weak law only needs pairwise independent tosses. The cost of using Chebyshev is the factor $1/\delta$: for $\delta = 10^{-6}$ it asks for 10^8 tosses. Independent tosses allow Chernoff, where δ enters only as $\ln(1/\delta)$. □

3.4 Random Walk on a Small Graph

For a graph with m edges, $P_{uv} = 1/\deg(u)$ if $u \sim v$ and 0 otherwise, $p_{t+1} = p_t P$ and $\pi_v = \deg(v)/2m$. The walk converges to π from every start exactly when the graph is connected and not bipartite; a graph is bipartite exactly when it has no odd cycle.

Problem 3.7 (A triangle with a tail)**.** Vertices $1, 2, 3$ form a triangle and vertex 4 is joined to vertex 3.

(a) Write P.

(b) Find π and verify it.

(c) Does the walk converge?

(d) Starting at vertex 4, compute p_1, p_2, p_3.

Solution. *Scribes:* Lemma 17, p. 20; Theorem 18, p. 22; Theorem 19, p. 22.

(a) The degrees are $2, 2, 3, 1$, so

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Every row sums to 1.

(b) There are $m = 4$ edges, so $\pi = \left(\frac{2}{8}, \frac{2}{8}, \frac{3}{8}, \frac{1}{8}\right) = \left(\frac{1}{4}, \frac{1}{4}, \frac{3}{8}, \frac{1}{8}\right)$. Verify $\pi P = \pi$ entry by entry:

- Vertex 1: $\pi_2 \cdot \frac{1}{2} + \pi_3 \cdot \frac{1}{3} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$.
- Vertex 2: $\pi_1 \cdot \frac{1}{2} + \pi_3 \cdot \frac{1}{3} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$.
- Vertex 3: $\pi_1 \cdot \frac{1}{2} + \pi_2 \cdot \frac{1}{2} + \pi_4 \cdot 1 = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$.
- Vertex 4: $\pi_3 \cdot \frac{1}{3} = \frac{1}{8}$.

(c) Yes. The graph is connected, and it contains a triangle (an odd cycle), so it is not bipartite.

(d) $p_0 = (0, 0, 0, 1)$.

- $p_1 = (0, 0, 1, 0)$: from 4 the walk must go to 3.
- $p_2 = \left(\frac{1}{3}, \frac{1}{3}, 0, \frac{1}{3}\right)$: from 3 each neighbour has probability $\frac{1}{3}$.
- $p_3 = \left(\frac{1}{3} \cdot \frac{1}{2},\ \frac{1}{3} \cdot \frac{1}{2},\ \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 1,\ 0\right) = \left(\frac{1}{6}, \frac{1}{6}, \frac{2}{3}, 0\right)$.

Each p_t sums to 1, as it must (a row stochastic matrix preserves probability vectors). □

Problem 3.8 (A path on three vertices)**.** Consider the path $1 - 2 - 3$. Find π. Does the walk from vertex 1 converge?

Solution. *Scribes:* Theorem 18, p. 22; Theorem 19, p. 22.

1. The degrees are $1, 2, 1$ and $m = 2$, so $\pi = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right)$. Check: $\pi P = \left(\frac{1}{2} \cdot \frac{1}{2},\ \frac{1}{4} + \frac{1}{4},\ \frac{1}{2} \cdot \frac{1}{2}\right) = \pi$.
2. The graph is bipartite, with sides $\{1, 3\}$ and $\{2\}$.
3. From $p_0 = (1, 0, 0)$: $p_1 = (0, 1, 0)$, $p_2 = \left(\frac{1}{2}, 0, \frac{1}{2}\right)$, $p_3 = (0, 1, 0)$, and so on.
4. The distribution alternates forever between $(0, 1, 0)$ and $\left(\frac{1}{2}, 0, \frac{1}{2}\right)$. It does not converge, although π exists and satisfies $\pi P = \pi$.

Beyond the scribes (the eigenvalue proof is not in them): P has the eigenvalues 1, 0 and -1. The eigenvalue -1 is what a bipartite graph produces, and its contribution $(-1)^t$ never dies out. □

Problem 3.9 (Short questions)**.** Find the stationary distribution and say whether the walk converges for

(a) a 4-cycle and a 5-cycle,

(b) any d-regular graph on n vertices,

(c) a disconnected graph.

Solution. *Scribes:* Theorem 18, p. 22; Theorem 19, p. 22.

(a) In a 4-cycle every degree is 2, so π is uniform, $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$. An even cycle is bipartite, so the walk from a single vertex does not converge. A 5-cycle is an odd cycle, so the walk does converge, to the uniform distribution.

(b) $2m = nd$, so $\pi_v = d/(nd) = 1/n$: the stationary distribution is uniform.

(c) Each component has its own stationary distribution, so π is not unique and the limit depends on the start. □

Problem 3.10 (Rows, not columns)**.** The Sep 10 exercise uses that every row of P sums to 1. Show by an example that columns summing to 1 is not enough: find a matrix Q with nonnegative entries whose columns sum to 1, and a probability vector p, such that pQ is not a probability vector.

Solution. *Scribes:* Exercise 8, p. 21.

Take

$$Q = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad p = (1, 0).$$

Both columns sum to 1, but $pQ = (1, 1)$, whose entries sum to 2. In general the entries of pQ sum to $\sum_j \sum_i p_i Q_{ij} = \sum_i p_i \left(\sum_j Q_{ij}\right)$: this is 1 for every probability vector p exactly when every row sum $\sum_j Q_{ij}$ is 1. □

4 Week 4: Qubits, Unitaries, Entanglement, Rank

Lectures of September 15 and 17.

In the scribes	Result	Page
Exercise: entries of the column vector $\|00\rangle$	Exercise 9	23
Exercise: norm preserving iff unitary	Exercise 10	24
$HH = I$	Lemma 20	25
Matrix of CNOT	Lemma 21	26
XOR is not reversible	Lemma 22	26
Exercise: the Bell state is entangled	Exercise 11	27
Exercise: rank of the example matrix	Exercise 12	29
Exercise: row rank equals column rank	Exercise 13	29
Exercise: orthogonal implies independent	Exercise 14	30

4.1 Basis Vectors, Unitaries, Gates

Problem 4.1 (Which entry is 1?)**.** Write $|10\rangle$ as a column vector in $\mathbb{C}^4$, and say which entry of $|101\rangle$ in $\mathbb{C}^8$ is 1.

Solution. *Scribes:* Exercise 9, p. 23.

The basis states are ordered by the binary number they spell: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ are the entries $1, 2, 3, 4$. So $|10\rangle = |1\rangle \otimes |0\rangle = (0,1)^T \otimes (1,0)^T = (0,0,1,0)^T$. The string 101 is the number 5, so $|101\rangle$ has a 1 in entry $5 + 1 = 6$ and zeros elsewhere. □

Problem 4.2 (Is the matrix unitary?)**.** Decide for each matrix whether it is unitary.

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad W = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Solution. *Scribes:* Exercise 10, p. 24.

Compute $U^\dagger U$, where $\dagger$ means: transpose, and conjugate every entry.

$$U^\dagger U = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1-i^2 & i-i \\ -i+i & -i^2+1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

So U is unitary. Do not forget the conjugate: without it the diagonal would be $1 + i^2 = 0$.

V is not unitary: $V|1\rangle = |0\rangle + |1\rangle$ has norm $\sqrt{2}$, and a unitary matrix preserves norms (Sep 15). W is not unitary: $W|0\rangle = W|1\rangle = |+\rangle$, so two orthogonal states are sent to the same state; also $W(|0\rangle - |1\rangle) = 0$.

Quick test. A matrix is unitary exactly when its columns are orthonormal, with the complex inner product. □

Problem 4.3 (HXH)**.** Compute HXH and HZH, where $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Solution. *Scribes:* Lemma 20, p. 25.

$XH = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}$, so

$$HXH = \frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}2 & 0\\ 0 & -2\end{pmatrix} = Z.$$

Multiply $HXH = Z$ by H on both sides and use $HH = I$: $X = HZH$. So H swaps the bit flip and the phase flip. On states: X fixes $|+\rangle$ and negates $|-\rangle$, just as Z fixes $|0\rangle$ and negates $|1\rangle$. □

4.2 Entangled or Product

Write a two-qubit state as $a|00\rangle+b|01\rangle+c|10\rangle+d|11\rangle$. A product $(\alpha_0|0\rangle + \alpha_1|1\rangle)(\beta_0|0\rangle + \beta_1|1\rangle)$ has $a = \alpha_0\beta_0$, $b = \alpha_0\beta_1$, $c = \alpha_1\beta_0$, $d = \alpha_1\beta_1$, so $ad = \alpha_0\alpha_1\beta_0\beta_1 = bc$. If $ad \neq bc$ the state is entangled; this is the Bell state argument written once for all states. To show that a state is a product, write down the two factors.

Problem 4.4 (Five states)**.** Which of these states are entangled? Prove each answer.

(a) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

(b) $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$

(c) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$

(d) $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

(e) $\sqrt{1/3}\,|00\rangle + \sqrt{2/3}\,|01\rangle$

Solution. *Scribes:* Exercise 11, p. 27.

Compare ad with bc:

State	ad	bc	Answer
(a)	$1/4$	$1/4$	Product: $\|+\rangle\|+\rangle$
(b)	$-1/4$	$-1/4$	Product: $\|+\rangle\|-\rangle$
(c)	$-1/4$	$1/4$	Entangled
(d)	0	$-1/2$	Entangled
(e)	0	0	Product: $\|0\rangle\left(\sqrt{1/3}\,|0\rangle + \sqrt{2/3}\,|1\rangle\right)$

Full write-up for (c). Suppose the state equals $(\alpha_0|0\rangle + \alpha_1|1\rangle)(\beta_0|0\rangle + \beta_1|1\rangle)$. Then $\alpha_0\beta_0 = \frac{1}{2}$, $\alpha_0\beta_1 = \frac{1}{2}$, $\alpha_1\beta_0 = \frac{1}{2}$ and $\alpha_1\beta_1 = -\frac{1}{2}$. The product of the first and the last is $\alpha_0\alpha_1\beta_0\beta_1 = -\frac{1}{4}$. The product of the middle two is $\alpha_0\alpha_1\beta_0\beta_1 = +\frac{1}{4}$. This is a contradiction, so the state is entangled. □

Problem 4.5 (A small circuit)**.** Apply H to the first qubit and then CNOT (the first qubit is the control) to $|10\rangle$. Is the result entangled?

Solution. *Scribes:* Lemma 21, p. 26; Exercise 11, p. 27.

1. H on the first qubit: $|10\rangle$ becomes $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle)$.

2. CNOT flips the second qubit when the first is 1: $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$.

3. Test: $ad = -\frac{1}{2}$ and $bc = 0$, so the state is entangled.

The same circuit sends $|00\rangle$ to $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, $|01\rangle$ to $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ and $|11\rangle$ to $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$. □

4.3 Rank and Independence

Problem 4.6 (Rank of a 3×3 matrix)**.** Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}.$$

Find the column rank, a basis of $\mathrm{Im}(A)$, and the row rank, and write every row as a combination of two rows.

Solution. *Scribes:* Exercise 12, p. 29; Exercise 13, p. 29.

Columns. $c_1 = (1, 2, 1)^T$, $c_2 = (2, 4, 0)^T$, $c_3 = (3, 6, 1)^T = c_1 + c_2$. The columns c_1 and c_2 are independent: if $c_2 = \lambda c_1$, the third entries give $\lambda = 0$, but $c_2 \neq 0$. So $\mathrm{rank}_c(A) = 2$ and $\mathrm{Im}(A) = \mathrm{span}(c_1, c_2)$.

Rows. $r_1 = [1\ 2\ 3]$, $r_2 = [2\ 4\ 6]$, $r_3 = [1\ 0\ 1]$. Here $r_2 = 2r_1$, and r_1, r_3 are independent. So $\mathrm{rank}_r(A) = 2$, equal to the column rank as the exercise says. Every row in terms of r_1, r_3: $r_1 = 1 \cdot r_1 + 0 \cdot r_3$, $r_2 = 2r_1 + 0 \cdot r_3$, $r_3 = 0 \cdot r_1 + 1 \cdot r_3$. □

Problem 4.7 (Orthogonal does not imply independent over $\{0,1\}$)**.** Over $\{0,1\}^n$ with $x \cdot y = \sum_i x_i y_i \bmod 2$, is it still true that a nonzero v orthogonal to independent $v_1, \ldots, v_k$ is independent of them? Which step of the proof breaks?

Solution. *Scribes:* Exercise 14, p. 30.

No. In $\{0,1\}^2$ take $v_1 = 11$ and $v = 11$. Then $v \cdot v_1 = 1 + 1 = 0 \bmod 2$, so v is nonzero and orthogonal to v_1, but $v = v_1$ is not independent of it.

The proof takes $c\,v + \sum_i c_i v_i = 0$, takes the inner product with v to get $c\|v\|^2 = 0$, and concludes $c = 0$ from $\|v\|^2 > 0$. Over $\{0,1\}$ the analogue of $\|v\|^2$ is $v \cdot v$, the number of 1s of v mod 2. It is 0 for every v with an even number of 1s, so the step $c = 0$ fails.

Looking ahead to week 6. In Simon's algorithm every measured y satisfies $s \cdot y = 0$, and s itself may be one of them (when s has an even number of 1s, $s \cdot s = 0$). So independence of the y's cannot come from orthogonality; it is a probability statement, proved separately in week 6 (Exercise 20, p. 42). □

5 Week 5: U_f, Phase Kickback, Deutsch and Deutsch–Jozsa

Lecture of September 24. There is no scribe for September 22.

In the scribes	Result	Page
The oracle $U_f\|x\rangle\|y\rangle = \|x\rangle\|y \oplus f(x)\rangle$	Figure 11	31
Exercise: $H\frac{1}{\sqrt{2}}(\|0\rangle + (-1)^a\|1\rangle)$	Exercise 15	32
Exercise: Deutsch's algorithm gives $f(0) \oplus f(1)$	Exercise 16	32
Walsh–Hadamard transform	Lemma 25	33
Exercise: Deutsch–Jozsa outcome 0^n iff constant	Exercise 17	34

Two facts do all the work:

$$U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle, \qquad H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} (-1)^{x\cdot y}|y\rangle.$$

To run an algorithm on a given f, write the state after every gate.

Problem 5.1 (Why do we need U_f?)**.** Let $f\colon \{0,1\}^n \to \{0,1\}$.

(a) Why can no quantum circuit map $|x\rangle$ to $|f(x)\rangle$?

(b) Show that $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ is unitary.

(c) How do we read $f(x)$ from it, and where does the sign $(-1)^{f(x)}$ come from?

Solution. *Scribes:* Lemma 22, p. 26; Figure 11, p. 31.

(a) Every quantum gate is unitary, so it preserves norms and can be undone (Sep 15). The map $x \mapsto f(x)$ is not one-to-one as soon as two inputs have the same value, and with more than two inputs that always happens. In vectors: if $f(x) = f(x')$ with $x \neq x'$, a linear map T with $T|x\rangle = |f(x)\rangle$ sends $|x\rangle - |x'\rangle$ to 0. That vector has norm $\sqrt{2}$, so T does not preserve norms. This is the Sep 15 argument for XOR, where 00 and 11 both give 0.

(b) U_f sends the basis state $|x\rangle|y\rangle$ to the basis state $|x\rangle|y \oplus f(x)\rangle$. Applying it twice gives $y \oplus f(x) \oplus f(x) = y$, so $U_f^2 = I$. So U_f permutes the 2^{n+1} basis states, its matrix is a permutation matrix, and a permutation matrix is unitary (the Sep 15 argument for CNOT). The trick is that x is kept: from $(x, y \oplus f(x))$ we can recover y, so nothing is lost.

(c) With $y = 0$: $U_f|x\rangle|0\rangle = |x\rangle|f(x)\rangle$, so measuring the last qubit gives $f(x)$. With the last qubit in $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

$$U_f|x\rangle|-\rangle = |x\rangle \cdot \frac{1}{\sqrt{2}}(|f(x)\rangle - |1 \oplus f(x)\rangle).$$

If $f(x) = 0$ this is $|x\rangle|-\rangle$, and if $f(x) = 1$ it is $-|x\rangle|-\rangle$. So $U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$, the phase kickback used in Problem 5.4.

CNOT is the smallest example: it is U_f for the one-bit function $f(x) = x$. □

Problem 5.2 (Deutsch on $f(x) = 1 - x$)**.** Run Deutsch's algorithm on $f(0) = 1$, $f(1) = 0$, writing the state after every gate.

Solution. *Scribes:* Exercise 15, p. 32; Exercise 16, p. 32.

1. Start in $|0\rangle|1\rangle$. After $H \otimes H$: $|+\rangle|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|-\rangle$.
2. The query multiplies $|x\rangle$ by $(-1)^{f(x)}$: $\frac{1}{\sqrt{2}}(-|0\rangle + |1\rangle)|-\rangle = -|-\rangle|-\rangle$.
3. H on the first qubit: $H|-\rangle = |1\rangle$, so the state is $-|1\rangle|-\rangle$.
4. The first qubit is measured as 1 with probability 1. Indeed $f(0) \oplus f(1) = 1$: f is balanced.

The sign -1 is global and does not change the measurement. □

Problem 5.3 ($H^{\otimes 2}$ on a basis state)**.** Compute $H^{\otimes 2}|10\rangle$ with the Walsh–Hadamard formula, and check it as a product.

Solution. *Scribes:* Lemma 25, p. 33.

With $x = 10$, $x \cdot y = y_1$, so

$$H^{\otimes 2}|10\rangle = \frac{1}{2}\sum_y (-1)^{y_1}|y\rangle = \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle).$$

As a product: $H|1\rangle \otimes H|0\rangle = |-\rangle|+\rangle = \frac{1}{2}(|0\rangle - |1\rangle)(|0\rangle + |1\rangle)$, which multiplies out to the same four terms. □

Problem 5.4 (Deutsch–Jozsa with $n = 2$)**.** Let $f(x_1x_2) = x_1 \oplus x_2$. Run the Deutsch–Jozsa algorithm on f. What is measured, and what does the algorithm conclude?

Solution. *Scribes:* Exercise 17, p. 34; Lemma 25, p. 33.

1. Start in $|00\rangle|1\rangle$. Hadamards on all three qubits give $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|-\rangle$.
2. The query multiplies each $|x\rangle$ by $(-1)^{f(x)}$. Here $f(00) = 0$, $f(01) = 1$, $f(10) = 1$, $f(11) = 0$, so the state is $\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)|-\rangle$.
3. Hadamards on the first two qubits. The amplitude of $|y\rangle$ is $\frac{1}{4}\sum_x (-1)^{f(x)+x\cdot y}$.
4. For $y = 00$ this is $\frac{1}{4}(1 - 1 - 1 + 1) = 0$. For $y = 11$ we have $x \cdot y = x_1 \oplus x_2 = f(x)$, so every term is $+1$ and the amplitude is 1.
5. The measurement gives 11 with probability 1. The outcome is not 00, so f is balanced.

One quantum query was used. A deterministic classical algorithm needs $2^{n-1} + 1 = 3$ queries.

Check with a constant function. For $f \equiv 1$ the state after the query is $-\frac{1}{2}\sum_x |x\rangle = -|+\rangle|+\rangle$, and the Hadamards turn it into $-|00\rangle$. The measurement gives 00, so f is constant. The global sign -1 does not affect the measurement. □

Problem 5.5 (Deutsch–Jozsa needs $N/2 + 1$ deterministic queries)**.** Prove that a deterministic classical algorithm for the Deutsch–Jozsa problem needs $2^{n-1} + 1$ queries.

Solution. *Scribes:* Exercise 17, p. 34.

1. Let $N = 2^n$. The adversary answers 0 to every query.
2. Suppose the algorithm stops after at most $N/2$ queries. At least $N/2$ inputs are unqueried.
3. Completion 1: $f = 0$ everywhere, which is constant.
4. Completion 2: $f = 1$ on exactly $N/2$ of the unqueried inputs and 0 elsewhere, which is balanced.
5. Both satisfy the promise and agree with every answer, so the algorithm is wrong on one. □

Problem 5.6 (Is Deutsch–Jozsa an exponential speedup?)**.** The quantum algorithm uses 1 query and a deterministic one needs $N/2 + 1$. Is this an exponential quantum advantage?

Solution. *Scribes:* Exercise 17, p. 34.

Only against deterministic (or zero-error) algorithms. A randomized algorithm queries α random inputs and answers "balanced" if two answers differ. It is always right on a constant f. On a balanced f all α answers agree with probability $2 \cdot \left(\frac{1}{2}\right)^{\alpha} = \left(\frac{1}{2}\right)^{\alpha-1}$. With $\alpha = 11$ queries the error is below $1/1000$, for every n. So against bounded-error randomized algorithms the gap is 1 query against a constant. □

6 Week 6: Bernstein–Vazirani and Simon

Lectures of September 29 and October 1.

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Sum of signs $\sum_x (-1)^{x \cdot y}$	Claim 26	35
Bernstein–Vazirani	Theorem 27	35
One run of Simon's circuit	Theorem 28	37
Simon's algorithm	Theorem 29	37
Exercise: deterministic $\mathcal{O}(\sqrt{2^n})$ algorithm	Exercise 18	38
Randomized upper bound	Theorem 30	39
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Yao's min-max principle, easy direction	Lemma 32	40
Few queries rarely collide	Lemma 33	40
Classical query complexity $\Theta(\sqrt{2^n})$	Corollary 34	41
Exercise: Simon's circuit when f is 1-to-1	Exercise 19	42
Exercise: random vectors are independent w.p. $\geqslant 1/4$	Exercise 20	42

6.1 Running the Algorithms

Problem 6.1 (Bernstein–Vazirani with $n = 3$)**.** Let $f(x) = s \cdot x \bmod 2$ with the hidden string $s = 101$, so $f(x) = x_1 \oplus x_3$.

(a) How does a classical algorithm find s?

(b) Run the quantum algorithm.

Solution. *Scribes:* Theorem 27, p. 35; Claim 26, p. 35.

(a) Query the unit vectors: $f(100) = 1$, $f(010) = 0$, $f(001) = 1$. The answers are the bits of s, so $n = 3$ queries. Fewer cannot work, since each answer is one bit and s has n bits.

(b) After the first Hadamards and the query, the first register is

$$\frac{1}{\sqrt{8}} \sum_x (-1)^{s \cdot x} |x\rangle = \frac{1}{\sqrt{8}} (|000\rangle - |001\rangle + |010\rangle - |011\rangle - |100\rangle + |101\rangle - |110\rangle + |111\rangle).$$

After the second Hadamards the amplitude of $|y\rangle$ is

$$\frac{1}{8} \sum_x (-1)^{s \cdot x + y \cdot x} = \frac{1}{8} \sum_x (-1)^{(s \oplus y) \cdot x} = \begin{cases} 1 & \text{if } y = s, \\ 0 & \text{if } y \neq s, \end{cases}$$

by the sum-of-signs claim (Sep 29): the sum is 2^n if $s \oplus y = 000$ and 0 otherwise. So the measurement gives $y = 101 = s$ with probability 1, after one query. □

Problem 6.2 (Simon with $n = 3$)**.** The function $f\colon \{0,1\}^3 \to \{0,1\}^3$ is given by the table below. Run Simon's algorithm and find s.

x	000	001	010	011	100	101	110	111
$f(x)$	101	010	000	110	000	110	101	010

Solution. *Scribes:* Theorem 28, p. 37; Theorem 29, p. 37; Exercise 20, p. 42.

1. After the Hadamards and the query, the state is $\frac{1}{\sqrt{8}}\sum_x |x\rangle|f(x)\rangle$.

2. Measure the second register. Suppose the outcome is 010. The inputs with $f(x) = 010$ are 001 and 111, so the first register is $\frac{1}{\sqrt{2}}(|001\rangle + |111\rangle)$.

3. Apply $H^{\otimes 3}$. The amplitude of $|y\rangle$ is

$$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{8}}\Big((-1)^{001\cdot y} + (-1)^{111\cdot y}\Big) = \frac{1}{4}(-1)^{001\cdot y}\Big(1 + (-1)^{110\cdot y}\Big).$$

4. This is $\pm\frac{1}{2}$ if $110 \cdot y = 0$ and 0 if $110 \cdot y = 1$. So y is one of $000, 001, 110, 111$, each with probability 1/4. The state is $\frac{1}{2}(|000\rangle - |001\rangle + |110\rangle - |111\rangle)$.

5. Every outcome satisfies $y_1 \oplus y_2 = 0$. The same holds for any other outcome in step 2, because every pair $\{x, x \oplus s\}$ has the same difference.

6. Run the circuit $n - 1 = 2$ times. Suppose the outcomes are $y = 111$ and $y = 001$. They give the equations $s_1 \oplus s_2 \oplus s_3 = 0$ and $s_3 = 0$.

7. Solve: $s_3 = 0$ and $s_1 = s_2$. The solutions are 000 and 110, so the nonzero candidate is $s' = 110$.

8. Check with two classical queries: $f(000) = 101 = f(110)$. So f is 2-to-1 and $s = 110$.

What can go wrong. If the two outcomes are dependent (for example 000 and 111, or 001 twice), the system has more than two solutions and we run again. Two outcomes are independent with probability $\frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$: the first must be nonzero (3 of 4), and the second must be outside $\{000, y_1\}$ (2 of 4). This matches the bound of at least 1/4 from Oct 1.

If f were 1-to-1. Step 2 would leave a single $|x\rangle$, every y would have probability 1/8, and the check in step 8 would fail, so the algorithm would answer $s = 000$. □

Problem 6.3 (Simon's circuit for a 1-to-1 f, $n = 2$)**.** Let $f\colon \{0,1\}^2 \to \{0,1\}^2$ be 1-to-1. After the second register is measured with outcome $f(x_0)$, what is the state, and with what probability is each y measured?

Solution. *Scribes:* Exercise 19, p. 42; Lemma 25, p. 33.

Only x_0 has the value $f(x_0)$, so the first register is the single basis state $|x_0\rangle$. Then $H^{\otimes 2}|x_0\rangle = \frac{1}{2}\sum_y (-1)^{x_0\cdot y}|y\rangle$, so each of the 4 strings y has probability $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$, and these add up to 1. For general n each y has probability $1/2^n$. Unlike the 2-to-1 case, no y is ruled out. □

Problem 6.4 (Independence for $n = 4$)**.** Simon's algorithm with $n = 4$ collects y_1, y_2, y_3. Find the probability that they are linearly independent if f is 2-to-1, and if f is 1-to-1. How many attempts are expected in the 2-to-1 case?

Solution. *Scribes:* Exercise 20, p. 42.

2-*to*-1. The y's are uniform in $V = \{y \mid s \cdot y = 0\}$, of dimension $d = 3$. Part 1 of the exercise gives

$$\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{8}\right) = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{7}{8} = \frac{21}{64} \approx 0.33 \geqslant \frac{1}{4}.$$

Each factor is the probability that the next vector is outside the span of the earlier ones: the span of $i-1$ independent vectors has 2^{i-1} of the 2^3 elements of V.

1-*to*-1. Now $V = \{0,1\}^4$, $d = 4$, and we need $d - 1 = 3$ independent vectors. Part 2 gives $\frac{3}{4} \cdot \frac{7}{8} \cdot \frac{15}{16} = \frac{315}{512} \approx 0.62 \geqslant \frac{1}{2}$.

Attempts. Each attempt succeeds with probability 21/64, independently, so the number of attempts is geometric with mean $64/21 \approx 3.05$ (Theorem 5, p. 7). □

Problem 6.5 (Deterministic Simon for $n = 4$)**.** Run the deterministic algorithm of the Sep 29 exercise for $n = 4$.

(a) Which inputs does it query, and how many?

(b) Which two queries collide if $s = 0110$? If $s = 1100$?

(c) How many queries for $n = 20$, against the obvious deterministic algorithm and the quantum one?

Solution. *Scribes:* Exercise 18, p. 38.

The algorithm splits x into its first $m = \lceil n/2 \rceil$ bits u and its last $n - m$ bits v, and queries every string of the form $(u, 0 \ldots 0)$ and every string of the form $(0 \ldots 0, v)$.

(a) Here $m = 2$. The strings $u00$ are $0000, 0100, 1000, 1100$. The strings $00v$ are $0000, 0001, 0010, 0011$. The string 0000 is in both, so there are $4 + 4 - 1 = 7$ queries.

(b) $s = 0110$ has $u = 01$ and $v = 10$. The queries 0100 and 0010 have XOR $0110 = s$, so $f(0100) = f(0010)$. $s = 1100$ has $u = 11$ and $v = 00$. The queries 1100 and 0000 have XOR $1100 = s$, so $f(1100) = f(0000)$.

In general every $s = (u, v) \neq 0000$ is the XOR of the two queried strings $(u, 00)$ and $(00, v)$, and these are different because $s \neq 0000$. So a collision always appears, and by the promise any collision x, x' has $x \oplus x' = s$. If no two answers agree, f is one-to-one and $s = 0000$.

(c) The count is $2^m + 2^{n-m} - 1$. For $n = 20$ that is $1024 + 1024 - 1 = 2047$. Querying inputs until two answers agree needs $2^{19} + 1 = 524{,}289$ queries in the worst case. Simon's algorithm makes $n + 1 = 21$.

So the deterministic count is about $2\sqrt{N}$, the same order as the randomized lower bound of Problem 6.7, and the quantum count is about $\log N$. □

6.2 Classical Lower Bounds and the Separation

Problem 6.6 (The same for Monte Carlo algorithms)**.** Show that a randomized search algorithm with success probability at least 2/3 on every input needs at least $2n/3 - 1$ queries.

Solution. *Scribes:* Lemma 32, p. 40; Theorem 1, p. 3.

1. Use the same distribution μ. Take a deterministic algorithm with T queries, in the fixed order $q_1, \ldots, q_T$ while the answers are "no".
2. It finds x if x is in one of these T slots: probability T/n.
3. Otherwise it has seen T answers of "no" and outputs one fixed slot. That slot is the

right one with probability at most $1/n$ overall.

4. So it succeeds with probability at most $(T+1)/n$ under μ.

5. If $T+1 < 2n/3$, every deterministic algorithm with T queries errs with probability more than $1/3$ under μ. By Yao's principle (the form proved on Oct 1) every randomized algorithm with T queries errs with probability more than $1/3$ on some input. □

Problem 6.7 (Simon's problem, in outline)**.** State the hard distribution for Simon's problem and the key estimate. Evaluate it for $n = 10$ and 16 queries.

Solution. *Scribes:* Theorem 31, p. 40; Lemma 33, p. 40.

1. Distribution: s is uniform among the $N-1$ nonzero strings, and f is a uniformly random 2-to-1 function with period s.

2. An algorithm learns s only from a collision, two queries x and $x \oplus s$.

3. With k distinct queries the probability of a collision is at most $k(k-1)/N$, since there are $k(k-1)/2$ pairs and each pair rules in one value of s out of at least $N/2$.

4. Without a collision, s is still uniform over at least $N/2$ strings, so a guess is right with probability at most $2/N$.

5. The success probability is at most $(k(k-1)+2)/N$.

For $n = 10$, $N = 1024$ and $k = 16 = \sqrt{N}/2$: at most $(240+2)/1024 \approx 0.24$. So $\sqrt{N}/2$ queries are not enough for success probability $2/3$, and the classical complexity is $\Omega\left(\sqrt{2^n}\right)$. □

Problem 6.8 (Why Simon is different)**.** Explain why Simon's problem gives a real exponential separation.

Solution. *Scribes:* Theorem 31, p. 40; Corollary 34, p. 41.

The Oct 1 lower bound is for randomized algorithms with bounded error: success probability $2/3$ needs more than $\sqrt{N}/2$ queries. The quantum algorithm needs $\mathcal{O}(n)$ queries. So the gap is $\sqrt{2^n}$ against n even when the classical algorithm may use randomness and may err. A quantum advantage has to be shown against randomized algorithms, because they are the strongest classical model (Aug 25). □

7 Week 7: Grover's Algorithm and Amplitude Amplification

Lectures of October 6 and 8.

In the scribes	Result	Page
Inversion about the mean	Lemma 37	45
One Grover iteration	Lemma 38	46
Exercise: G is the rotation $R_{2\theta}$	Exercise 21	47
Exercise: $D = 2\|h\rangle\langle h\| - I$	Exercise 22	48
Two reflections make a rotation	Claim 39	49
The state after k iterations	Corollary 40	49
Exercise: M^2 is the rotation by 2θ	Exercise 23	50
Exercise: number of iterations, success $\geqslant 1 - a/N$	Exercise 24	50
Exercise: random k, success $\geqslant 3/8$	Exercise 25	50
Exercise: amplitude amplification	Exercise 26	52

7.1 Grover with Given N and a

N is the number of inputs, a the number of solutions and k the number of iterations:

$$\sin\theta = \sqrt{\frac{a}{N}}, \qquad \mathbb{P}(\text{success after } k \text{ iterations}) = \sin^2((2k+1)\theta).$$

The scribe takes k to be an integer closest to $\frac{\pi}{4\theta} - \frac{1}{2}$, and proves success probability at least $1 - a/N$. Except at a tie this is the same as $k = \lfloor \pi/(4\theta) \rfloor$, which the solutions below use. Each iteration adds 2θ to the angle, and we want $(2k+1)\theta$ as close to 90° as possible.

Problem 7.1 ($N = 4$, $a = 1$)**.** Find θ, the best k and the success probability. Then verify by tracking the four amplitudes.

Solution. *Scribes:* Lemma 37, p. 45; Exercise 24, p. 50.

1. $\sin\theta = \sqrt{1/4} = \frac{1}{2}$, so $\theta = 30^\circ = \pi/6$.
2. $k = \lfloor \pi/(4\theta) \rfloor = \lfloor 1.5 \rfloor = 1$.
3. The success probability is $\sin^2(3 \cdot 30^\circ) = \sin^2 90^\circ = 1$. One query finds the solution with certainty; classically 3 queries are needed in the worst case.

Amplitude check. Start with $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$. The phase flip Z_f negates the solution: $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right)$, with the solution last. The mean is $\frac{1}{4}\left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} - \frac{1}{2}\right) = \frac{1}{4}$. Inversion about the mean sends each amplitude α to $2 \cdot \frac{1}{4} - \alpha$, which gives $(0, 0, 0, 1)$. □

Problem 7.2 (Inversion about the mean for $N = 8$, $a = 1$)**.** Track the eight amplitudes through one Grover iteration and compare with $\sin^2(3\theta)$.

Solution. *Scribes:* Lemma 37, p. 45; Corollary 40, p. 49.

1. Start: every amplitude is $1/\sqrt{8}$.
2. Z_f negates the solution: seven amplitudes $1/\sqrt{8}$ and one $-1/\sqrt{8}$.

3. The mean is $\frac{1}{8} \cdot \frac{7-1}{\sqrt{8}} = \frac{3}{4\sqrt{8}}$.

4. Each amplitude α becomes $2 \cdot \text{mean} - \alpha$. Non-solutions: $\frac{6}{4\sqrt{8}} - \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{8}}$. Solution: $\frac{6}{4\sqrt{8}} + \frac{1}{\sqrt{8}} = \frac{5}{2\sqrt{8}}$.

5. Success probability $\frac{25}{4 \cdot 8} = \frac{25}{32} \approx 0.781$. Check of the norm: $7 \cdot \frac{1}{32} + \frac{25}{32} = 1$.

With $\sin\theta = 1/\sqrt{8}$: $\sin 3\theta = 3\sin\theta - 4\sin^3\theta = \frac{3}{\sqrt{8}} - \frac{4}{8\sqrt{8}} = \frac{5}{2\sqrt{8}}$, the same amplitude. □

Problem 7.3 ($N = 8$, $a = 1$, and overshooting)**.** Find the best k, and the success probability for $k = 1, 2, 3$.

Solution. *Scribes:* Corollary 40, p. 49; Exercise 24, p. 50.

1. $\sin\theta = 1/\sqrt{8} \approx 0.3536$, so $\theta \approx 0.3614$ rad $\approx 20.7°$.

2. $k = \lfloor \pi/(4 \cdot 0.3614) \rfloor = \lfloor 2.17 \rfloor = 2$.

3. Plug in:

k	Angle $(2k+1)\theta$	Success probability
0	20.7°	$1/8 = 0.125$
1	62.1°	$25/32 \approx 0.781$
2	103.5°	$121/128 \approx 0.945$
3	144.9°	$169/512 \approx 0.330$

The best is $k = 2$ with probability about 0.945. A third iteration rotates past 90° and the probability drops to 0.33: more iterations are not better. □

Problem 7.4 ($N = 8$, $a = 2$)**.** Same N, two solutions: find θ, k and the success probability.

Solution. *Scribes:* Corollary 40, p. 49; Exercise 24, p. 50.

$\sin\theta = \sqrt{2/8} = \frac{1}{2}$, so $\theta = 30°$, $k = 1$ and the success probability is $\sin^2 90° = 1$. Only a/N matters, so this is the same as $N = 4$, $a = 1$. Whenever $a = N/4$, one iteration succeeds with certainty. □

Problem 7.5 (Using the wrong a)**.** Let $N = 64$. We run the number of iterations that is right for $a = 1$, but in fact $a = 4$. What is the success probability? What should k have been?

Solution. *Scribes:* Corollary 40, p. 49; Exercise 24, p. 50.

1. For $a = 1$: $\sin\theta = 1/8$, $\theta \approx 0.1253$ and $k = \lfloor 6.27 \rfloor = 6$. With $a = 1$ this would succeed with probability $\sin^2(13 \cdot 0.1253) \approx 0.997$.

2. With $a = 4$: $\sin\theta = \sqrt{4/64} = \frac{1}{4}$ and $\theta \approx 0.2527$. After 6 iterations the angle is

$13 \cdot 0.2527 \approx 3.285$ rad $\approx 188°$, and $\sin^2 \approx 0.02$.

3. The right choice is $k = \lfloor \pi/(4 \cdot 0.2527) \rfloor = \lfloor 3.11 \rfloor = 3$, with success probability $\sin^2(7 \cdot 0.2527) \approx 0.961$.

So with the wrong a the algorithm fails 98% of the time. This is why an unknown a needs a random k (Oct 8): choose k uniformly from $\left\{1, \ldots, \lfloor 2\sqrt{N} \rfloor\right\}$ and the success probability is at least $3/8$ for every a with $1 \leqslant a \leqslant N/2$ (Problem 7.7). □

Problem 7.6 ($a = N/2$ and $a > N/2$)**.** What does Grover's algorithm do when half or more of the inputs are solutions?

Solution. *Scribes:* Corollary 40, p. 49.

For $a = N/2$: $\theta = 45°$, and the angles $45°, 135°, 225°, \ldots$ all have $\sin^2 = \frac{1}{2}$. Iterating never helps. For $a > N/2$: $\theta > 45°$, so $k = \lfloor \pi/(4\theta) \rfloor = 0$, and measuring the uniform state directly succeeds with probability $a/N > \frac{1}{2}$. In both cases just sample a random x and check it with one query. □

Problem 7.7 (Grover when a is unknown)**.** Let $N = 64$, and let the number of solutions a be unknown, with $1 \leqslant a \leqslant 32$.

(a) Why not just run 6 iterations?

(b) State the algorithm of the Oct 8 exercise and its guarantee.

(c) Compute its success probability for $a = 1$ and for $a = 4$.

(d) How do we make the failure probability small?

Solution. *Scribes:* Exercise 25, p. 50.

(a) Six iterations is right for $a = 1$ (success 0.997), but for $a = 4$ it rotates past the target and succeeds with probability 0.02 (Problem 7.5). The right k depends on a.

(b) Let $K = \lfloor 2\sqrt{N} \rfloor = 16$. Choose k uniformly from $\{1, \ldots, 16\}$, make k Grover iterations on the uniform state and measure. The exercise proves

$$\mathbb{P}(\text{success}) = \frac{1}{K} \sum_{k=1}^{K} \sin^2((2k+1)\theta) = \frac{1}{2} - \frac{\sin(4(K+1)\theta) - \sin 4\theta}{4K \sin 2\theta} \geqslant \frac{1}{2} - \frac{1}{4K \sin 2\theta} \geqslant \frac{3}{8}.$$

(c) For $a = 1$: $\sin\theta = 1/8$ and $\theta \approx 0.1253$. Then $\sin 2\theta \approx 0.2480$, $\sin 4\theta \approx 0.4806$ and $\sin(68\theta) = \sin(8.522) \approx 0.7849$. The probability is

$$\frac{1}{2} - \frac{0.7849 - 0.4806}{64 \cdot 0.2480} \approx \frac{1}{2} - 0.019 = 0.481.$$

For $a = 4$: $\sin\theta = \frac{1}{4}$ and $\theta \approx 0.2527$. Then $\sin 2\theta \approx 0.4841$, $\sin 4\theta \approx 0.8472$ and $\sin(68\theta) = \sin(17.182) \approx -0.9953$. The probability is

$$\frac{1}{2} - \frac{-0.9953 - 0.8472}{64 \cdot 0.4841} \approx \frac{1}{2} + 0.059 = 0.559.$$

Both are above $3/8 = 0.375$. The reason it works: averaged over many values of k, $\sin^2$ of the angle is about $\frac{1}{2}$, whatever θ is. The last inequality of (b) holds because

$K \sin 2\theta \geqslant \sqrt{2N} \cdot \sqrt{2a/N} = 2\sqrt{a} \geqslant 2$, so the subtracted term is at most $1/8$. It uses $a \leqslant N/2$.

(d) Check the measured x with one query to f. If it is not a solution, repeat with a new random k. After r rounds the failure probability is at most $\left(\frac{5}{8}\right)^r$, which is below 1% for $r = 10$.

Each round costs at most $16 = 2\sqrt{N}$ queries, so the algorithm is still $\mathcal{O}\left(\sqrt{N}\right)$ without knowing a. □

7.2 Structure of the Iteration

Problem 7.8 (The rotation for $N = 4$, $a = 1$)**.** In the basis $(|B\rangle, |A\rangle)$, write the matrix G of one Grover iteration and the matrix M of the rotation by θ. Check that $M^2 = G$, and compute $G|h\rangle$.

Solution. *Scribes:* Exercise 21, p. 47; Exercise 23, p. 50.

Here $a = 1$, $b = 3$, $\sin\theta = \frac{1}{2}$, $\theta = 30°$.

1. G has entries $\frac{b-a}{N} = \frac{1}{2}$ and $\frac{2\sqrt{ab}}{N} = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$:
$$G = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = R_{60°}.$$
2. M has entries $\sqrt{b/N} = \frac{\sqrt{3}}{2} = \cos 30°$ and $\sqrt{a/N} = \frac{1}{2} = \sin 30°$, so $M = R_{30°}$.
3. $M^2 = \begin{pmatrix} \frac{3}{4} - \frac{1}{4} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} & -\frac{1}{4} + \frac{3}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = G.$
4. $|h\rangle = \cos 30°|B\rangle + \sin 30°|A\rangle$ has coordinates $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$, and $G|h\rangle = \left(\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4},\ \frac{3}{4} + \frac{1}{4}\right) = (0, 1)$, that is, $G|h\rangle = |A\rangle$.

The angle goes from 30° to 90° in one iteration, which is why Problem 7.1 succeeds with certainty. □

Problem 7.9 (D is a reflection)**.** Let $D = 2|h\rangle\langle h| - I$ with $\langle h|h\rangle = 1$. Show that D is unitary, $D^2 = I$, $D|h\rangle = |h\rangle$, and $D|v\rangle = -|v\rangle$ for every $|v\rangle$ orthogonal to $|h\rangle$.

Solution. *Scribes:* Exercise 22, p. 48.

$D|h\rangle = 2|h\rangle\langle h|h\rangle - |h\rangle = |h\rangle$, and if $\langle h|v\rangle = 0$ then $D|v\rangle = 2|h\rangle\langle h|v\rangle - |v\rangle = -|v\rangle$. Next,

$$D^2 = 4|h\rangle\langle h|h\rangle\langle h| - 4|h\rangle\langle h| + I = I.$$

Also $D^\dagger = 2|h\rangle\langle h| - I = D$, because $(|h\rangle\langle h|)^\dagger = |h\rangle\langle h|$. So $D^\dagger D = D^2 = I$ and D is unitary. Geometrically D keeps the $|h\rangle$ direction and flips everything orthogonal to it: a reflection about $|h\rangle$. □

7.3 Amplitude Amplification

For a measurement-free algorithm $\mathcal{A}$ that succeeds with probability p: $\sin\theta = \sqrt{p}$, $Q = -\mathcal{A}\,Z_0\,\mathcal{A}^{-1}Z_f$, success after k rounds $\sin^2((2k+1)\theta)$, and $k = \lfloor \pi/(4\theta) \rfloor$. The cost is $2k+1$ applications of $\mathcal{A}$ or $\mathcal{A}^{-1}$: one to prepare the state and two per round.

Problem 7.10 ($p = 1/100$)**.** A quantum algorithm $\mathcal{A}$ outputs a good answer with probability $1/100$, and we can check an answer with f. How many runs of $\mathcal{A}$ are needed classically, and how many with amplitude amplification?

Solution. *Scribes:* Exercise 26, p. 52.

1. Classically: run, check, repeat. The number of runs is geometric with mean $1/p = 100$.
2. $\sin\theta = \sqrt{1/100} = 0.1$, so $\theta \approx 0.1002$.
3. $k = \lfloor \pi/(4 \cdot 0.1002) \rfloor = \lfloor 7.84 \rfloor = 7$.
4. The success probability is $\sin^2(15 \cdot 0.1002) = \sin^2(1.503) \approx 0.995$.
5. Cost: $2 \cdot 7 + 1 = 15$ applications of $\mathcal{A}$ or its inverse, and 7 queries to f.

So 15 applications instead of 100. In general the cost is at most $\frac{\pi}{2\sqrt{p}} + 1 = \mathcal{O}(1/\sqrt{p})$ against $1/p$. □

Problem 7.11 ($p = 1/4$)**.** Answer the same question for $p = 1/4$.

Solution. *Scribes:* Exercise 26, p. 52.

$\sin\theta = \frac{1}{2}$, $\theta = 30°$, $k = 1$, and the success probability is $\sin^2 90° = 1$. Three applications of $\mathcal{A}$ or its inverse give a good answer with certainty. Classically the expected number of runs is 4, and no fixed number of runs gives certainty. □

Problem 7.12 (Short questions)**.**

(a) Why must $\mathcal{A}$ contain no measurement?

(b) How is Grover's algorithm a special case?

(c) What if p is unknown?

Solution. *Scribes:* Exercise 26, p. 52.

(a) The operator Q uses the inverse of $\mathcal{A}$. A unitary circuit is inverted by running it backwards with each gate inverted. A measurement is not unitary and cannot be undone.

(b) Take $\mathcal{A} = H^{\otimes n}$. Then $\mathcal{A}|0^n\rangle$ is the uniform state, $p = a/N$, and $Q = -H^{\otimes n}Z_0H^{\otimes n}Z_f$ is the Grover iteration.

(c) The formula for k needs p. The theorem in the scribe assumes that p is known; the unknown case was not covered. □

7.4 Comparing the Models

Problem 7.13 (The numbers for $n = 20$)**.** Fill in the query counts for $n = 20$, $N = 1{,}048{,}576$.

Solution. *Scribes:* Corollary 34, p. 41; Theorem 27, p. 35; Exercise 17, p. 34.

The classical count for Simon lies between the lower bound of Oct 1 and the deterministic algorithm of Sep 29.

Problem	Classical	Quantum
Deutsch–Jozsa, no error	524,289	1
Bernstein–Vazirani	20	1
Simon	more than 512 ($\sqrt{N}/2$); about 1024 suffice up to a constant	21 per attempt: 19 quantum and 2 classical
Search, one solution	1,048,575 deterministic	804 Grover iterations: $\lfloor \frac{\pi}{4} \cdot 1024 \rfloor$

□

Problem 7.14 (One-line reasons)**.** Give a one-line reason for each statement.

(a) Search is only a quadratic speedup.

(b) Bernstein–Vazirani is n against 1.

(c) Simon and collisions have the same square root.

Solution. *Scribes:* Theorem 27, p. 35; Exercise 3, p. 14.

(a) Grover uses $\mathcal{O}\left(\sqrt{N}\right)$ queries and the scribe notes that $\Omega\left(\sqrt{N}\right)$ quantum queries are necessary, so there is no exponential gain for unstructured search.

(b) Each classical answer is one bit and s has n bits. One quantum query returns all n bits because the Hadamard transform turns the sign pattern $(-1)^{s \cdot x}$ into $|s\rangle$.

(c) Classically s is found only by a collision, and k queries make about $k^2/2$ pairs, so k must reach about $\sqrt{N}$. This is the birthday paradox of Sep 3. □